

CS 506: Introduction to Quantum Computing

Spectral Theorem and Functions of Operators

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1 Overview

This lecture introduces the spectral theorem for normal operators. It begins with a simple illustration using the Pauli- X gate, derives its eigenvalues and orthonormal eigenvectors, and then uses spectral decomposition to define functions of operators. The final section reviews valid quantum states and gives a normalized complex-amplitude example.

2 Illustration of the Spectral Theorem

Consider the Pauli- X gate, which acts as a bit flip:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

To find its eigenvalues and eigenvectors, solve

$$\begin{aligned} X \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} &= \lambda \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}. \\ \implies \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} &= \lambda \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \end{aligned}$$

Matrix multiplication gives

$$\begin{pmatrix} x_2 \\ x_1 \end{pmatrix} = \begin{pmatrix} \lambda x_1 \\ \lambda x_2 \end{pmatrix},$$

so

$$x_2 = \lambda x_1, \quad x_1 = \lambda x_2.$$

Substituting the first relation into the second yields

$$x_1 = \lambda^2 x_1.$$

For a nonzero eigenvector, $x_1 \neq 0$, and therefore

$$\lambda^2 = 1, \quad \lambda \in \{+1, -1\}.$$

2.1 Orthonormal Eigenvectors

For $\lambda = 1$, we have $x_1 = x_2$. Normalization requires

$$\left\| \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \right\|^2 = |x_1|^2 + |x_2|^2 = 1.$$

Choosing real amplitudes gives

$$x_1 = x_2 = \frac{1}{\sqrt{2}}, \quad |+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle.$$

For $\lambda = -1$, we have $x_1 = -x_2$. A normalized eigenvector is

$$|-\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle.$$

Thus the eigenvalue equations are

$$X|+\rangle = |+\rangle, \quad X|-\rangle = -|-\rangle.$$

The vectors $|+\rangle$ and $|-\rangle$ are orthonormal and form a basis of $\mathbb{C}^2$.

2.2 Spectral Decomposition of the Pauli- X Gate

According to the Spectral Theorem, a Hermitian operator can be written as a weighted sum of the outer products of its normalized eigenvectors:

$$T = \sum_i \lambda_i |v_i\rangle\langle v_i|,$$

where λ_i is the eigenvalue corresponding to the eigenvector $|v_i\rangle$.

The Pauli- X operator has eigenvalue $+1$ with eigenvector $|+\rangle$ and eigenvalue -1 with eigenvector $|-\rangle$. Therefore, its spectral decomposition is

$$\begin{aligned} X &= (+1)|+\rangle\langle +| + (-1)|-\rangle\langle -| \\ &= |+\rangle\langle +| - |-\rangle\langle -|. \end{aligned}$$

This can be verified directly:

$$\begin{aligned} |+\rangle\langle +| &= \frac{1}{2}(|0\rangle + |1\rangle)(\langle 0| + \langle 1|), \\ |-\rangle\langle -| &= \frac{1}{2}(|0\rangle - |1\rangle)(\langle 0| - \langle 1|). \end{aligned}$$

Subtracting the two outer products and simplifying, we obtain

$$\begin{aligned} |+\rangle\langle +| - |-\rangle\langle -| &= \frac{1}{2}(|0\rangle\langle 0| + |0\rangle\langle 1| + |1\rangle\langle 0| + |1\rangle\langle 1|) \\ &\quad - \frac{1}{2}(|0\rangle\langle 0| - |0\rangle\langle 1| - |1\rangle\langle 0| + |1\rangle\langle 1|) \\ &= \frac{1}{2}[(|0\rangle\langle 0| - |0\rangle\langle 0|) \\ &\quad + (|0\rangle\langle 1| - (-|0\rangle\langle 1|)) \\ &\quad + (|1\rangle\langle 0| - (-|1\rangle\langle 0|)) \\ &\quad + (|1\rangle\langle 1| - |1\rangle\langle 1|)] \\ &= \frac{1}{2}[2|0\rangle\langle 1| + 2|1\rangle\langle 0|] \\ &= |0\rangle\langle 1| + |1\rangle\langle 0| \\ &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = X. \end{aligned}$$

3 Functions and Power-Series Expansions

Generally, a scalar function may be written as

$$f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \cdots = \sum_{i=0}^{\infty} a_i x^i.$$

Many operator functions are understood through familiar power-series expansions. For example,

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots = \sum_{i=0}^{\infty} \frac{x^i}{i!}.$$

The Taylor–Maclaurin expansions also give

$$\begin{aligned} e^{ix} &= 1 + ix + \frac{(ix)^2}{2!} + \frac{(ix)^3}{3!} + \frac{(ix)^4}{4!} + \frac{(ix)^5}{5!} + \cdots \\ &= 1 + ix - \frac{x^2}{2!} - i\frac{x^3}{3!} + \frac{x^4}{4!} + i\frac{x^5}{5!} - \cdots \\ &= \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots\right) + i\left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots\right) \end{aligned}$$

From Euler’s identity we know:

$$e^{ix} = \cos x + i \sin x.$$

Comparing the real parts we get,

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$$

4 Normal Operators and the Spectral Theorem

Let T be a normal operator represented by an $n \times n$ matrix. Normality means

$$TT^\dagger = T^\dagger T.$$

Hermitian and unitary matrices are important examples of normal matrices. The spectral theorem states that a normal operator has eigenvalues $T_1, T_2, \dots, T_n$ and a corresponding orthonormal eigenbasis

$$|T_1\rangle, |T_2\rangle, \dots, |T_n\rangle.$$

Therefore,

$$T |T_i\rangle = T_i |T_i\rangle \quad (i = 1, \dots, n).$$

4.1 Spectral Decomposition

Using the orthonormal eigenbasis, the operator can be written as

$$T = \sum_{i=1}^n T_i |T_i\rangle\langle T_i|.$$

Each outer product $|T_i\rangle\langle T_i|$ is the projector onto the eigenspace associated with T_i . Indeed,

$$T |T_j\rangle = \sum_{i=1}^n T_i |T_i\rangle \langle T_i | T_j \rangle = T_j |T_j\rangle,$$

because orthonormality gives $\langle T_i | T_j \rangle = \delta_{ij}$.

4.2 Powers of an Operator

Starting from the spectral decomposition

$$T = \lambda_1 |T_1\rangle\langle T_1| + \lambda_2 |T_2\rangle\langle T_2| + \cdots + \lambda_n |T_n\rangle\langle T_n|,$$

we calculate T^2 as follows:

$$\begin{aligned} T^2 &= T \cdot T \\ &= \left(\sum_{i=1}^n \lambda_i |T_i\rangle\langle T_i| \right) \left(\sum_{j=1}^n \lambda_j |T_j\rangle\langle T_j| \right) \\ &= \sum_{i=1}^n \sum_{j=1}^n \lambda_i \lambda_j |T_i\rangle \langle T_i | T_j \rangle \langle T_j|. \end{aligned}$$

Because the eigenvectors form an orthonormal basis,

$$\langle T_i | T_j \rangle = \delta_{ij}.$$

Thus, all terms for which $i \neq j$ vanish, and we obtain

$$\begin{aligned} T^2 &= \sum_{i=1}^n \lambda_i^2 |T_i\rangle\langle T_i| \\ &= \lambda_1^2 |T_1\rangle\langle T_1| + \lambda_2^2 |T_2\rangle\langle T_2| + \cdots + \lambda_n^2 |T_n\rangle\langle T_n|. \end{aligned}$$

Similarly,

$$\begin{aligned} T^m &= \lambda_1^m |T_1\rangle\langle T_1| + \lambda_2^m |T_2\rangle\langle T_2| + \cdots + \lambda_n^m |T_n\rangle\langle T_n| \\ &= \sum_{i=1}^n \lambda_i^m |T_i\rangle\langle T_i|. \end{aligned}$$

5 Functions of Normal Operators

Suppose the scalar function f has the power-series representation

$$f(x) = \sum_{m=0}^{\infty} a_m x^m.$$

Replacing the scalar variable by the operator T gives

$$\begin{aligned} f(T) &= \sum_{m=0}^{\infty} a_m T^m \\ &= \sum_{m=0}^{\infty} a_m \left(\sum_i T_i^m |T_i\rangle\langle T_i| \right) \\ &= \sum_i \left(\sum_{m=0}^{\infty} a_m T_i^m \right) |T_i\rangle\langle T_i| \\ &= \sum_i f(T_i) |T_i\rangle\langle T_i|. \end{aligned}$$

Thus, applying f to a normal operator leaves its eigenvectors unchanged and applies f only to its eigenvalues.

5.1 The Exponential of the Pauli- X Gate

Using the spectral decomposition of X ,

$$X = |+\rangle\langle+| - |-\rangle\langle-|,$$

we obtain

$$e^X = e^1 |+\rangle\langle+| + e^{-1} |-\rangle\langle-|.$$

In the computational basis,

$$\begin{aligned} e^X &= \frac{e}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + \frac{e^{-1}}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \\ &= \begin{pmatrix} \frac{e+e^{-1}}{2} & \frac{e-e^{-1}}{2} \\ \frac{e-e^{-1}}{2} & \frac{e+e^{-1}}{2} \end{pmatrix}. \end{aligned}$$

6 State Space Postulate

A state vector $|\psi\rangle$ is a valid quantum state when it is a complex column vector with unit norm:

$$|\psi\rangle \in \mathbb{C}^n, \quad \langle\psi|\psi\rangle = \|\psi\|^2 = 1.$$

The zero vector is therefore not a quantum state. The normalization condition ensures that the measurement probabilities sum to one.

For a one-qubit state

$$|\psi\rangle = \alpha_0 |0\rangle + \alpha_1 |1\rangle,$$

the amplitudes must satisfy

$$|\alpha_0|^2 + |\alpha_1|^2 = 1.$$

Examples include

$$|0\rangle, \quad |1\rangle, \quad \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle, \quad \frac{1}{\sqrt{2}}|0\rangle + \frac{i}{\sqrt{2}}|1\rangle.$$

6.1 Complex-Amplitude Example

Let

$$|\psi\rangle = \left(\frac{1}{\sqrt{6}} + i\sqrt{\frac{2}{6}}\right)|0\rangle + \left(\frac{1}{\sqrt{2}} + i\sqrt{\frac{5}{12}}\right)|1\rangle.$$

The squared magnitudes of the two amplitudes are

$$\begin{aligned} \left|\frac{1}{\sqrt{6}} + i\sqrt{\frac{2}{6}}\right|^2 &= \frac{1}{6} + \frac{2}{6} = \frac{1}{2}, \\ \left|\frac{1}{\sqrt{2}} + i\sqrt{\frac{5}{12}}\right|^2 &= \frac{1}{2} + \frac{5}{12} = \frac{11}{12}. \end{aligned}$$

Their sum is

$$\frac{1}{2} + \frac{11}{12} = \frac{17}{12} \neq 1.$$

Hence, as written, this vector is not normalized.

7 Key Takeaways

- A normal operator has an orthonormal eigenbasis and can be expressed as a weighted sum of eigenprojectors.
- Powers of a normal operator are obtained by raising its eigenvalues to the corresponding powers.
- For a suitable function f , the functional calculus is

$$f(T) = \sum_i f(T_i) |T_i\rangle\langle T_i|.$$

- Quantum state vectors must be normalized; the squared magnitudes of their amplitudes sum to one.